

CALL_FLOW

Call Flow

Call hierarchies for implemented major flows. Paths are relative to BankOne/src/main/java/com/bankone/ unless noted.

Login

```
AuthenticationController.login()
    ↓
AuthenticationService.login()
    ↓
AuthenticationManager.authenticate()
    ↓
CustomUserDetailsService.loadUserByUsername()
    ↓
UserRepository / UserRoleRepository
    ↓
JwtService.generateToken()
    ↓
(LoginAttemptService on failure)
```

Frontend: AuthService.login() → store token → navigate /app/dashboard.

Get current profile

```
AuthenticationController.me()
    ↓
AuthenticationService.getCurrentUserProfile()
    ↓
SecurityContext → UserRepository / roles
```

Create customer (optional first account)

```
CustomerController.createCustomer()
    ↓
CustomerServiceImpl.createCustomer()
    ↓
CustomerRepository.save()
    ↓
(if accountType present)
AccountService.openAccount() → see Open Account
```

Open account

```
AccountController.openAccount()
    ↓
AccountServiceImpl.openAccount()
```

↓
CustomerRepository.findById()
↓
AccountPolicyRepository.findByAccountTypeAndCurrencyCodeAndActiveTrue()
↓
validate opening deposit vs policy
↓
AccountRepository.getNextOrdinal()
↓
AccountNumberGenerator.generate()
↓
AccountRepository.save()

Frontend (Add Current):

CustomerDetail.addCurrentAccount()
↓
AccountPolicyService.getActivePolicy()
↓
OpeningDepositDialog (if required)
↓
AccountService.create / openAccount POST

Get active account policy

AccountPolicyController.getActivePolicy()
↓
AccountPolicyServiceImpl.getActivePolicy()
↓
AccountPolicyRepository.findByAccountTypeAndCurrencyCodeAndActiveTrue()

Deposit

AccountController.deposit()
↓
AccountServiceImpl.deposit()
↓
AccountRepository.findById()
↓
assert ACTIVE + amount > 0
↓
increment availableBalance, ledgerBalance, creditCount, timestamps
↓
AccountRepository.save()

Frontend: AccountList / deposit dialog → POST /accounts/{id}/deposit.

Search accounts

AccountController.searchAccounts()
↓
AccountServiceImpl.searchAccounts()
↓
AccountSpecification.matching()
↓
AccountRepository.findAll(spec, pageable)

Search customers

```
CustomerController.getCustomers()  
    ↓  
CustomerServiceImpl.searchCustomers()  
    ↓  
CustomerSpecification.containsText()  
    ↓  
CustomerRepository.findAll(spec, pageable)
```

Update account status

```
AccountController.updateAccountStatus()  
    ↓  
AccountServiceImpl.updateAccountStatus()  
    ↓  
AccountRepository.findById() / save()
```

Employee list / create (ADMIN)

```
UserController.*  
    ↓  
UserService.*  
    ↓  
UserRepository / UserRoleRepository / RoleRepository
```

Dashboard summary

```
DashboardController.getDashboard()  
    ↓  
DashboardServiceImpl  
    ↓  
CustomerRepository.count() / AccountRepository / UserRepository  
    ↓  
todayTransactionCount = 0 (stub)
```